

System Modeling and Analysis

1.1 Aim

To model and analyze the step response of a second-order RC circuit using Simulink and an STM32 Microcontroller.

1.2 List of Equipment

Table 1.1: List of Equipment

S.No.	Equipment	Value
1	Resistance	10k
2	Capacitance	47 μ F
3	STM32 Microcontroller	F47zi
4	Breadboard	1
5	5V Power Supply	1

1.3 Theory

1.3.1 Modeling Second-order RC circuit

Consider the second-order RC circuit shown in Figure 1.1. The transfer function of the circuit is given by,

$$H(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{1}{R_1 R_2 C_1 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_1 C_2)s + 1} \quad (1.1)$$

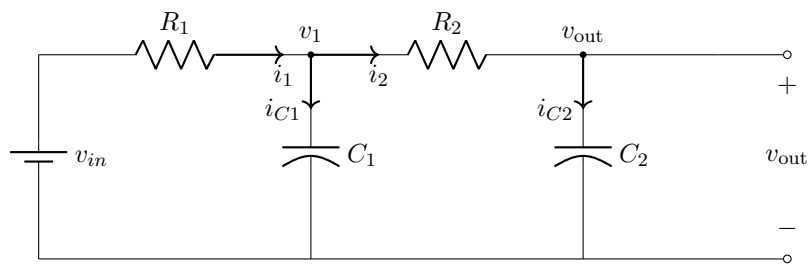


Figure 1.1: Cascaded second-order RC circuit with 5V input battery. Node voltages v_1 and v_{out} are shown. Currents i_1 and i_2 indicate the direction of current flow for node analysis.

The transfer function can be derived using Kirchhoff's Current Law (KCL) and the Laplace transform. The circuit consists of two resistors R_1 and R_2 and two capacitors C_1 and C_2 .

Step 1: Write KCL equations

At node between R_1 and C_1 (let voltage be $v_1(t)$):

$$i_1 = i_{C1} + i_2$$

$$\frac{v_{\text{in}}(t) - v_1(t)}{R_1} = C_1 \dot{v}_1(t) + \frac{v_1(t) - v_{\text{out}}(t)}{R_2}$$

At node between R_2 and C_2 (output node):

$$i_2 = i_{C2}$$

$$\frac{v_1(t) - v_{\text{out}}(t)}{R_2} = C_2 \dot{v}_{\text{out}}(t)$$

Step 2: Express $v_1(t)$ from node 2 equation

$$v_1(t) = v_{\text{out}}(t) + R_2 C_2 \dot{v}_{\text{out}}(t)$$

Step 3: Substitute into node 1 equation

$$\frac{v_{\text{in}}(t) - (v_{\text{out}}(t) + R_2 C_2 \dot{v}_{\text{out}}(t))}{R_1} = C_1 (\dot{v}_{\text{out}}(t) + R_2 C_2 \ddot{v}_{\text{out}}(t)) + \frac{v_{\text{out}}(t) + R_2 C_2 \dot{v}_{\text{out}}(t) - v_{\text{out}}(t)}{R_2}$$

Simplifying the equation gives:

$$\frac{v_{\text{in}}(t) - v_{\text{out}}(t) - R_2 C_2 \dot{v}_{\text{out}}(t)}{R_1} = C_1 (\dot{v}_{\text{out}}(t) + R_2 C_2 \ddot{v}_{\text{out}}(t)) + C_2 \dot{v}_{\text{out}}(t)$$

Step 4: Multiply both sides by R_1 and rearrange

$$v_{\text{in}}(t) - v_{\text{out}}(t) - R_2 C_2 \dot{v}_{\text{out}}(t) = R_1 C_1 \dot{v}_{\text{out}}(t) + R_1 C_1 R_2 C_2 \ddot{v}_{\text{out}}(t) + R_1 C_2 \dot{v}_{\text{out}}(t)$$

Group terms:

$$v_{\text{in}}(t) = v_{\text{out}}(t) + (R_2 C_2 + R_1 C_1 + R_1 C_2) \dot{v}_{\text{out}}(t) + R_1 R_2 C_1 C_2 \ddot{v}_{\text{out}}(t)$$

Step 5: Take Laplace transform (assuming zero initial conditions)

$$V_{\text{in}}(s) = V_{\text{out}}(s) [1 + s (R_2 C_2 + R_1 C_1 + R_1 C_2) + s^2 R_1 R_2 C_1 C_2]$$

Step 6: Transfer function

$$H(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{1}{R_1 R_2 C_1 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_1 C_2)s + 1}$$

1.4 Simulation with Simulink

To simulate the step response of the second-order RC circuit in Simulink, follow these steps:

1. Open Simulink and create a new model.
2. Drag and drop the following blocks into the model:
 - Step (from the Sources library)
 - Transfer Function (from the Continuous library)
 - Scope (from the Sinks library)
3. Configure the Step block:
 - Set the Step time to 0.
 - Set the Initial value to 0.
 - Set the Final value to 1.
4. Configure the Transfer Function block:
 - Set the Numerator coefficients to [1].

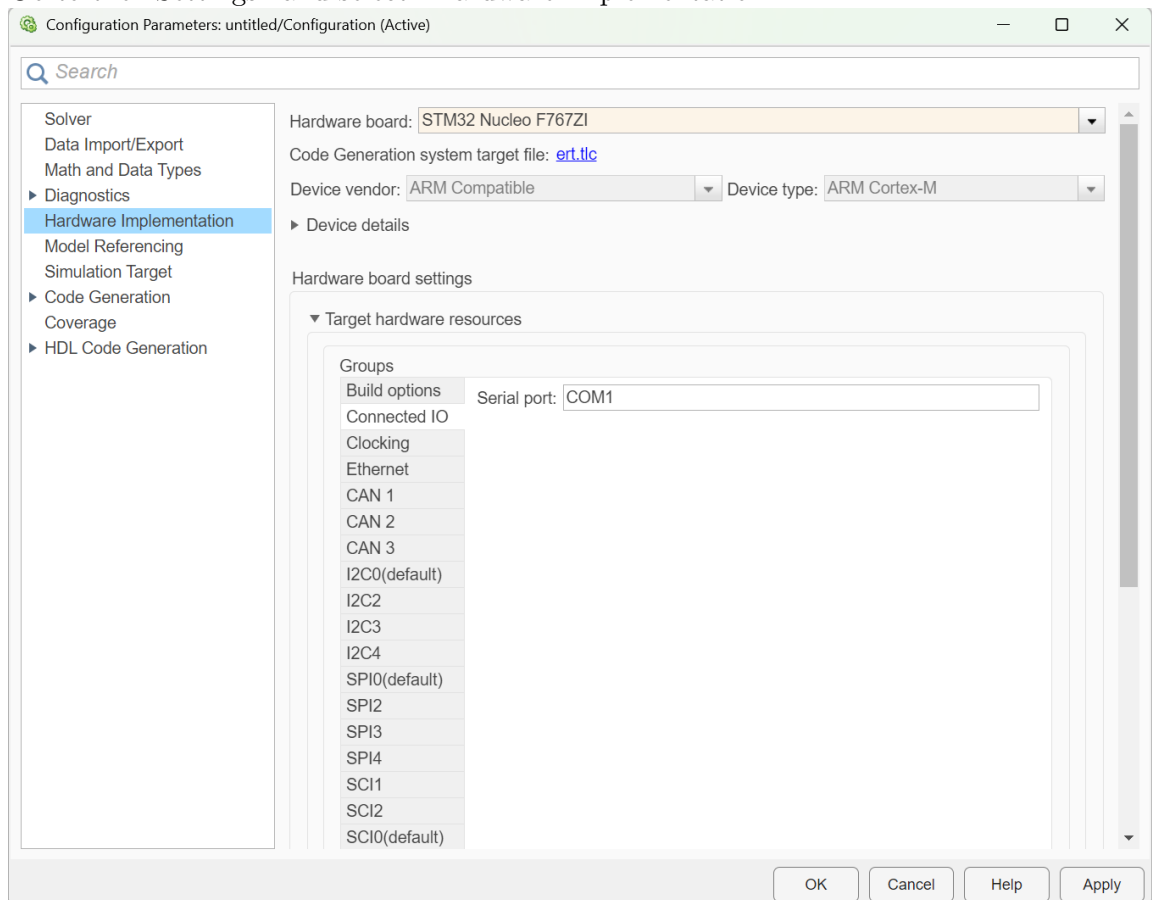
- Set the Denominator coefficients to $[R_1C_1R_2C_2, (R_2C_2 + R_1C_1 + R_1C_2), 1]$. Choose the appropriate values for R_1 , C_1 , R_2 , and C_2 .
5. Connect the blocks as follows:
 - Connect the output of the Step block to the input of the Transfer Function block.
 - Connect the output of the Transfer Function block to the input of the Scope block.
 - Enable Data logging on the Scope block to save the simulation data.
 6. Run the simulation.
 7. Observe and store the step response of the circuit on Simulation Data Inspector.

1.5 Hardware Implementation

The response of the system will be measured with the help of MATLAB and STM32 Microcontroller. For hardware implementation, please ensure that the “Simulink Coder Support Package for STMicroelectronics Nucleo Boards” is installed in your MATLAB environment. This package allows you to deploy Simulink models directly to STM32 microcontrollers.

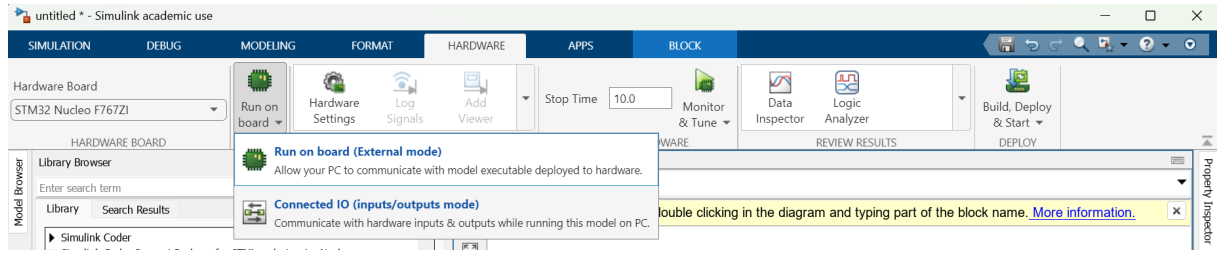
The following steps outline the process for hardware implementation:

1. Connect the STM32 Microcontroller to your computer and ensure it is recognized by MATLAB.
2. Open the Simulink model created in the previous section.
3. Configure the model for hardware deployment:
 - Go to the “Settings” and select “Hardware Implementation”.



- Choose “STMicroelectronics Nucleo” as the hardware board.
- Set the appropriate parameters for the STM32 board, such as the COM port, Baudrate and clock frequency.

- Save the model with a suitable name and ensure that the matlab file is saved in the current working directory.



4. Use the “Run on Board” or “Connected IO” option then click on “Monitor and Tune” in Simulink to compile and upload the model to the STM32 Microcontroller.
5. Once deployed, the STM32 will execute the model in real-time, and you can observe the step response on the connected hardware.
6. Use Simulink Data Inspector to visualize the step response data collected from the STM32 Microcontroller.
7. Optionally, you can use MATLAB scripts to further analyze the data collected from the STM32 Microcontroller.

Please refer the Matlab documentation for the STM32 Nucleo Boards Support from Embedded Coder [1].

1.6 Observation

Observation 1: For Ideal values

For the given values:

$$R_1 = 10 \text{ k}\Omega, \quad C_1 = 47 \text{ }\mu\text{F}, \quad R_2 = 10 \text{ k}\Omega, \quad C_2 = 47 \text{ }\mu\text{F}$$

Substitute these into the transfer function:

$$H(s) = \frac{1}{R_1 R_2 C_1 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_1 C_2)s + 1}$$

Calculate the coefficients:

$$R_1 C_1 = 10,000 \times 47 \times 10^{-6} = 0.47$$

$$R_2 C_2 = 0.47$$

$$R_1 C_2 = 0.47$$

$$R_1 R_2 C_1 C_2 = 10,000 \times 10,000 \times (47 \times 10^{-6})^2 = 100,000,000 \times 2.209 \times 10^{-9} \approx 0.221$$

So the denominator becomes:

$$0.221s^2 + (0.47 + 0.47 + 0.47)s + 1 = 0.221s^2 + 1.41s + 1$$

Final transfer function:

$$H(s) = \frac{1}{0.221s^2 + 1.41s + 1}$$

Now compare the step response obtained from the Simulink simulation with the theoretical transfer function and the hardware implementation results. The step response should match closely. Calculate the Integral Square Error (ISE) between the simulation and hardware implementation results to quantify the accuracy of the model.

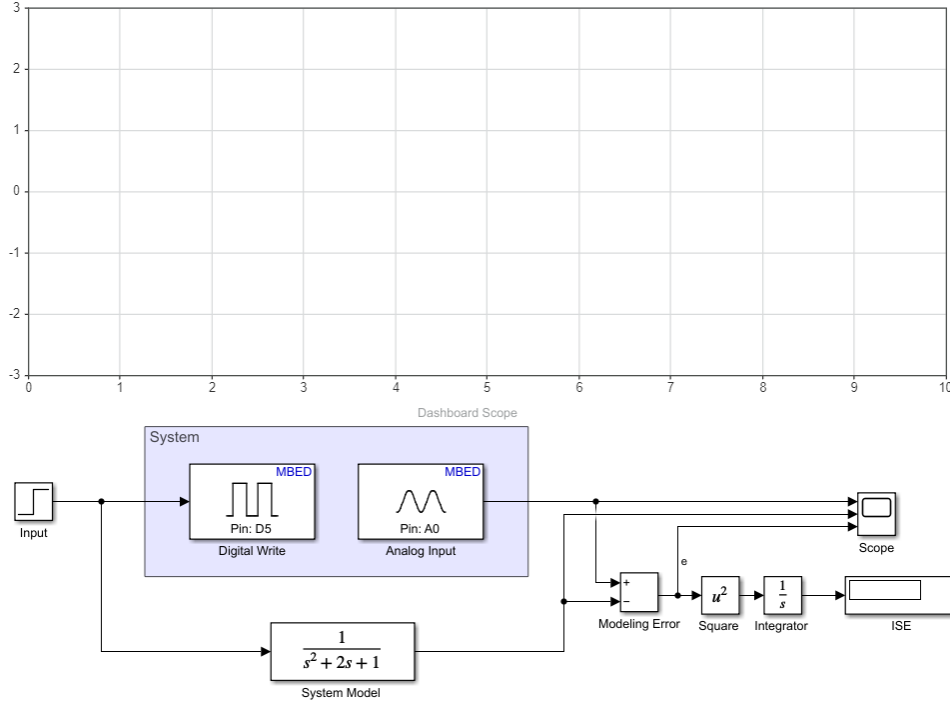


Figure 1.2: Simulink model for hardware implementation and Comparison with Simulation

Observation 2: For Measured values

Now let us consider the measured values from the comparison of Hardware implementation and simulation:

$$R_1 = 10\text{k}\Omega, \quad C_1 = 47\mu\text{F}, \quad R_2 = 10\text{k}\Omega, \quad C_2 = 47\mu\text{F}$$

The Simulink model for the hardware implementation is shown in Figure 1.2. Compare the Simulation results with the hardware implementation results. The step response should match closely.

1.6.1 Comparison of Modeling with rated value and Hardware Implementation

The step response obtained from the Simulink simulation and the hardware implementation using the STM32 Microcontroller should be compared. There should be a close match between the two responses, indicating that the theoretical model accurately represents the physical system. If there are discrepancies, they may arise from non-ideal components, parasitic effects, or measurement errors in the hardware setup. It is essential to analyze these differences to improve the model and understand the limitations of the theoretical assumptions.

Please list all the causes of the discrepancies observed in the step response between the simulation and hardware implementation. Also attach figure here.

1.6.2 Comparison of modeling with measured value with Hardware Implementation

Write the detailed comparison and attach the figure here.

1.7 Conclusion

In this experiment, we modeled and analyzed the step response of a second-order RC circuit using Simulink and an STM32 Microcontroller. The transfer function was derived, and the step response was simulated in Simulink. The hardware implementation was successfully carried out using the STM32 Microcontroller, allowing for real-time observation of the circuit's response. This experiment

demonstrates the integration of theoretical concepts with practical applications in circuit analysis and control systems.